

Sparse cumulative workload-state verifier

Every edge is a real release-to-deadline service opportunity; no imputed deadline or heuristic repair is used.

1 Arrival ledger

job j : release r_j
deadline d_j
energy E_j
region / class / site

$x_{j,t} = 0$ if $t \notin [r_j, d_j]$
immutable positive-energy join

edges



2 Cumulative state

$$y_t = y_{t-1} + x_t$$

$$y_t \leq A_t$$

$$y_{d_j} - y_{r_j-1} = E_j$$

release, conservation, deadline
capacity and cross-midnight state
are linear equalities/inequalities

LP



3 Global LP projection

$$\min ||p(x) - p^*||_1 + C(x)$$

HiGHS global optimum
validation selects risk reserve
convex hull keeps feasibility

exact job-flow witness (Exp. 14)
measured trace replay (Exp. 13)

certificate



4 Credit and payment gate

$$p_{\text{safe}} \leq p^{\{(\ell^*)\}}$$

pointwise false-credit envelope
 $V(p; \xi) = \max \text{dual affine}(\xi)$

q01/q10/q50/q90: finite samplewise cap
q99: fixed/flexible activation endpoint
N-1 security and signed value